

Low-cost gibberish screening: pygarble and DistilBERT on a complete published corpus

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Abstract

Local text screening requires a trade-off between detecting suspicious input, retaining meaningful text and inference cost. We compare five fixed pygarble configurations with a published DistilBERT gibberish classifier on the complete labelled release of a human-generated gibberish research corpus: 38 invented texts and 71 meaningful documents. Every normalized character is evaluated through 79,969 consecutive chunks, with English controls separated from other-language diagnostics. Under a fixed majority-chunk document rule, word lookup and two transformer label policies detect all 38 positive documents; pygarble’s English and extended profiles detect 13 and 20. Word lookup flags none of the 5,200 English control chunks. The transformer’s document-macro English false-positive rate is 48.47% when all non-clean classes trigger a flag, and 0.14% when only noise and word salad do. The transformer’s median single-thread CPU scoring latency is 1057 times word lookup’s on the fixed timing workload. These findings characterize invented-word detection in this collection, not general semantic understanding or deployment accuracy. We release pinned code, prediction-level artifacts, source-aware summaries and reproducibility checks.

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Introduction

A first-stage text filter may flag malformed or suspicious input before a more expensive processing step. Such a filter must balance detection against the cost of rejecting meaningful material. Character statistics and lexical coverage can be inexpensive, but historical spelling, names and specialist vocabulary can resemble noise to an English detector. A pretrained neural classifier offers another operating point, with additional inference and installation costs. Neither approach should be assumed to recognize meaning in every domain.

pygarble provides deterministic strategies, configurable ensembles and inspectable scores for local screening (Rao 2026). Its base installation has no runtime dependencies. Its separate PII, secrets and profanity module is outside this study. We compare its existing gibberish configurations with a publicly available DistilBERT classifier (Jindal, n.d.; Sanh et al. 2019), using a published linguistic research corpus (Gaskell and Bown 2022). Our questions are:

1. How do the fixed configurations compare on all released invented texts and all text from the English controls?
2. How sensitive is the comparison to the transformer’s binary label mapping and the rule for aggregating chunk decisions?
3. What false-flag behavior appears outside English, and what CPU and memory costs accompany each implementation?

The contribution is a reproducible empirical comparison, including negative results and inference costs. It is not a new classifier architecture. In particular, we test rather than assume that an ensemble improves on its individual strategies. The corpus’s limited source diversity prevents broad accuracy claims, even when every released character is processed.

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Corpus and coverage

Published source labels

Gaskell and Bown collected handwritten invented-language samples while studying statistical resemblance to the Voynich manuscript (Gaskell and Bown 2022). Their reported collection involved 42 volunteers and additional author samples. We evaluate all 38 transcript files in their released archive at GitHub revision d076a7d081f35098fa405928239595afd2e75927. We do not infer individual cohort membership or claim that the archive contains every sample from the original study.

We also include all 71 documents in the accompanying meaningful-text archive. These span 33 source language/variant labels, including historical and constructed languages. Four are explicitly English: the KJV New Testament, a modern NET Bible text, a historical Secreta Alberti transcription, and a Wikipedia-derived Voynich article. Those four provide the primary negative controls. The remaining 67 meaningful documents are a separate scope diagnostic; we retain their negative labels and do not redefine foreign-language text as gibberish. Voynichese has no established binary label and is not included. Thus, *complete corpus* refers to the entire released labelled collection.

All classifications come from the source collection. No new human annotation or label audit was performed. This includes inheriting a document’s label for short fragments, an assumption whose consequences we examine by excluding tails. The gibberish material carries the authors’ modified MIT notice and citation requirement; comparison documents retain their separate source terms. We publish hashes, normalized offsets and predictions, without redistributing comparison prose. No new participants were recruited.

Exhaustive chunking and dependence

We collapse whitespace to single spaces, retaining other characters, then partition each normalized document into consecutive, nonoverlapping chunks of 400 Unicode characters. The final short chunk is retained. This produces 31,964,664 normalized characters and 79,969 chunks across 109 documents:

Source group	Documents	Chunks	Role
Human-generated gibberish	38	173	Positive class
English meaningful text	4	5,200	Primary negative controls
Other meaningful text	67	74,596	Separate language diagnostic

No normalized character is dropped, sampled away or silently truncated. A chunk can end within a word; the same rule applies to every method and class. There are 109 short final chunks. Exact-hash checking finds 0 duplicate chunk occurrences beyond their first occurrence; all remain in the coverage analysis. Document IDs and source metadata remain attached to every prediction. Correlated passages and translations are not independent examples.

An earlier exploratory study used one 400-character prefix per positive and 100 selected blocks per English document. The full run contains 438 of those 438 primary input hashes. It expands coverage but is not an independent replication. The preliminary outcomes were known when this extension was designed. Its protocol and scoring code were committed as 717ceee before the new predictions; this is a versioned prospective analysis plan for the extension, not a preregistered or previously untouched benchmark.

Methods

pygarble configurations

We evaluate the unchanged `english`, `english_extended` and `legacy` profiles, and the individual `word_lookup` and `entropy_based` strategies. Each uses its default threshold of 0.5 and one thread. Ensemble defaults use any voting. The legacy profile combines Markov-chain, log-likelihood-ratio and word-anomaly strategies. English adds encoding/control-character and keyboard signals; extended also adds pattern, local-anomaly and repetition checks. The separately evaluated `word_lookup` strategy is not a member of these profiles. It uses an embedded English word set, folds diacritics, tokenizes Latin letters and half-weights unknown title-cased words. Strategy applicability

is preserved: an insufficient-evidence result does not trigger a flag and counts as a miss if the inherited label is positive. Abstention counts are retained separately from accuracy statistics.

These configurations share code and resources; they are not five independent software systems. No package resource, threshold or default was altered after observing this corpus. An always-keep rule would have zero recall and zero false flags and serves as the trivial reference without an additional inference run.

Hugging Face benchmark and label mapping

The external benchmark is Madhur Jindal’s published AutoNLP gibberish detector (Jindal, n.d.), a 66,956,548-parameter DistilBERT sequence classifier (Sanh et al. 2019). We pin revision 76672dd7d357 (full commit in the asset manifest) and record hashes for all model and tokenizer assets. Its safetensors weights occupy 267,838,720 bytes. The model card declares an MIT license. Its named training dataset was not accessible through the unauthenticated public API, so fine-tuning overlap with the evaluation material is unknown. Neither the model card’s own validation accuracy nor its marketing claims are used as our results.

The supplied labels are *clean*, *mild gibberish*, *noise* and *word salad*. We fix two binary policies before scoring: **HF non-clean** flags any winning class except clean; **HF noise/salad** flags only noise or word salad. Both policies use the same probabilities and model, and both are reported regardless of outcome. The winner is the highest-probability class; thresholding the sum of non-clean probabilities at 0.5 would be a different decision rule. There is no fine-tuning or threshold selection on this corpus.

Inference is local and offline after verified asset downloads. The full run uses PyTorch 2.9.0, Transformers 4.53.3, full-precision CPU inference, evaluation mode, no gradients, scaled-dot-product attention, four intra-op threads, one inter-op thread and batches of eight. Tokenizer parallelism is disabled. Inputs are dynamically padded without truncation; the largest observed input has 348 tokens, within the 512-token limit. Both systems receive identical text. The transformer is a research benchmark dependency only; pygarble’s base package remains model-free.

Endpoints and statistical units

A document is flagged if **at least half of its chunks are flagged**, with ties flagged. This is an explicit study aggregation rule, not a native long-document interface of either detector. Primary results are positive-document recall and false flags among the four English documents. To expose behavior hidden by majority aggregation, we also report each document’s flagged-chunk fraction and average those fractions with equal document weight. We call these the positive mean flagged fraction and negative mean chunk FPR. All micro counts are available, but 79,969 chunks are not treated as 79,969 independent samples.

Sensitivity analyses report any-chunk document decisions and mean chunk rates excluding final short chunks. Any-chunk flags tend to increase with document length; comparing their counts without that context would be misleading. Other-language meaningful texts receive their own document and language summaries, without pooling them into the English headline.

The 95% Wilson intervals for positive-document recall are conditional on document independence, which unknown participant metadata may violate. A 2,000-replicate paired document bootstrap, seeded 20260926, describes recall differences from HF non-clean. We do not give passage-binomial confidence intervals for FPR: 5,200 English chunks still come from four documents, two of which are Bible translations. No multiple-testing significance claim is made. Artificial class balance also makes headline accuracy, precision and F1 poor substitutes for deployment evaluation.

Results

Detection and English false flags

Table 1 reports document decisions and equally weighted within-document chunk rates. The last two columns refer only to the four English meaningful sources.

Method	TP docs	95% CI (%)	Pos. mean	FP docs	Neg. mean
English	13/38	21.2–50.1	34.2%	0/4	1.21%
Extended	20/38	37.3–67.5	47.9%	0/4	1.89%
Legacy	13/38	21.2–50.1	32.1%	0/4	0.25%
Word lookup	38/38	90.8–100.0	100.0%	0/4	0.00%
Entropy	0/38	0.0–9.2	0.0%	0/4	0.00%
HF non-clean	38/38	90.8–100.0	100.0%	1/4	48.47%
HF noise/salad	38/38	90.8–100.0	99.6%	0/4	0.14%

Table 1. TP/FP refer to majority-chunk document decisions. Pos./neg. means average flagged-chunk fractions across positive/English negative documents. Recall intervals describe 38 documents under the stated independence assumption.

Word lookup and both HF policies flag all 38 positive documents. Word lookup and HF non-clean also flag all 173 positive chunks; HF noise/salad flags 172. Perfect observed document recall has a conditional Wilson interval of roughly 90.8–100%, and does not establish equivalence on a wider population. Default English and extended profiles detect 13 and 20 positive documents, respectively; these default configurations detect fewer positive documents than the separate word-lookup strategy. This is a comparison of configurations, not an ablation of a shared ensemble component. Entropy produces no flags at its default operating point.

The English controls show a substantial policy effect in the transformer. HF non-clean flags 1,494 of 5,200 chunks and a majority of the Secreta document. HF noise/salad flags six chunks and no control-document majority. Its macro FPR is 0.14%, compared with 48.47% for HF non-clean. The difference is entirely due to the treatment of the model’s winning *mild gibberish* class, not retraining or different inputs. Word lookup has zero observed false flags in these four controls; this is not a zero-FPR guarantee outside the collection.

Method	KJV	NET	Wiki	Secreta
English	15/2379	19/2419	2/125	5/277
Extended	22/2379	24/2419	3/125	9/277
Legacy	4/2379	1/2419	1/125	0/277
Word lookup	0/2379	0/2419	0/125	0/277
Entropy	0/2379	0/2419	0/125	0/277
HF non-clean	579/2379	584/2419	59/125	272/277
HF noise/salad	5/2379	0/2419	0/125	1/277

Table 2. False-flag counts / complete chunk counts for each English source. A majority-only table would obscure the considerable within-document errors of HF non-clean on KJV, NET and Wiki, even though those documents remain below the majority threshold.

Aggregation and short-fragment sensitivity

Method	Any TP	Any FP	Full TP	Full pos. mean	Full neg. mean
English	18/38	4/4	13/38	35.9%	1.21%
Extended	26/38	4/4	18/38	48.6%	1.90%
Legacy	15/38	3/4	13/38	33.3%	0.25%
Word lookup	38/38	0/4	38/38	100.0%	0.00%
Entropy	0/38	0/4	0/38	0.0%	0.00%

Method	Any TP	Any FP	Full TP	Full pos. mean	Full neg. mean
HF non-clean	38/38	4/4	38/38	100.0%	48.37%
HF noise/salad	38/38	2/4	38/38	100.0%	0.14%

Table 3. Any-chunk document decisions, tail-excluded majority positive detections (Full TP), and document-macro flagged fractions after excluding short tails. Full-width rates still describe correlated chunks. The alternative document rule and tail analysis are sensitivity checks, not additional independent experiments or grounds to select an optimal rule.

The primary comparison favors a simple lexical signal on invented-word text. This should not be read as evidence about grammatical but semantically incoherent sentences: the corpus was not designed to balance those types of error. The two transformer policies have identical document recall here, despite one differing positive-chunk decision and markedly different control behavior.

Other-language scope diagnostic

Method	Other-language FP documents	Mean chunk FPR
English	16/67	28.4%
Extended	18/67	32.1%
Legacy	16/67	25.1%
Word lookup	49/67	69.1%
Entropy	12/67	17.9%
HF non-clean	62/67	91.9%
HF noise/salad	52/67	75.8%

Table 4. False flags on the 67 meaningful documents outside the explicitly English controls. These are retained source negatives, not extra gibberish examples. Rates average documents equally and remain outside the primary English comparison.

Word lookup falsely flags 49 of the 67 other-language documents under the majority rule; HF non-clean and HF noise/salad flag 62 and 52. No saved pygarble evaluation reports an API-level insufficient-evidence status.

These results test scope, not multilingual competence. The word-lookup strategy returns a zero score when it finds no eligible Latin-letter words; zero flags therefore need not establish linguistic coverage. Conversely, unfamiliar vocabulary or script can trigger flags in other configurations. The diagnostic separately records API-level insufficient-evidence counts; that API status is not itself a guarantee of language coverage. Related Bible content, parallel translations and the small number of documents per language limit cross-language comparisons. Neither English detector should be presented as a language-independent arbiter of meaningfulness on this evidence.

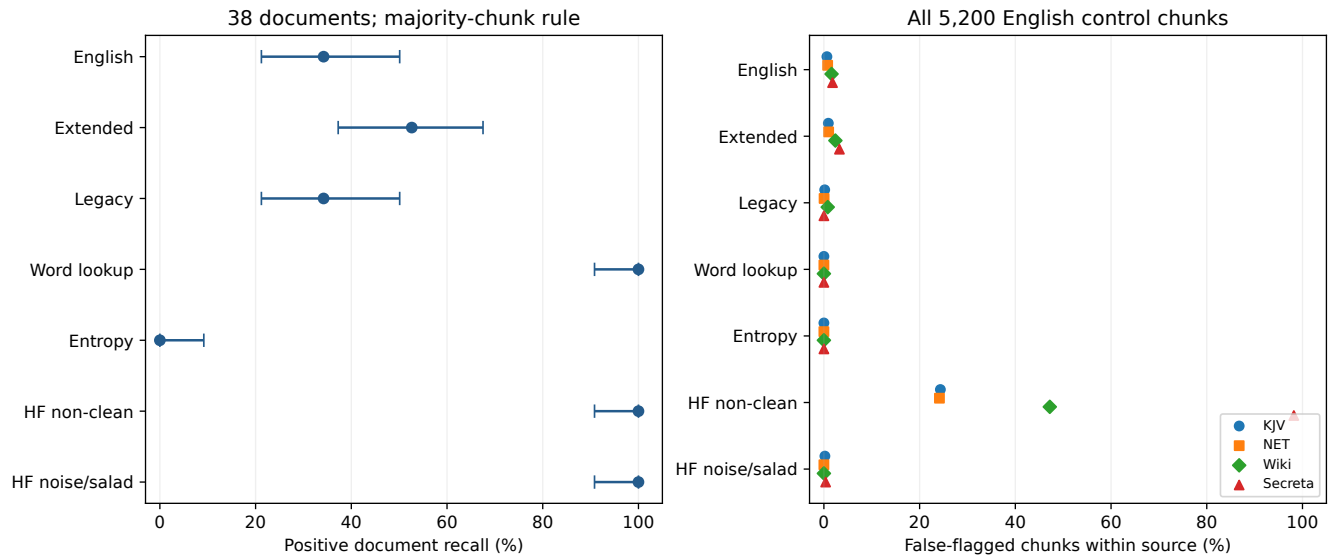


Figure 1: Complete English comparison. Left: majority-chunk recall over 38 positive documents, with conditional Wilson intervals. Right: complete within-source English false-flag rates.

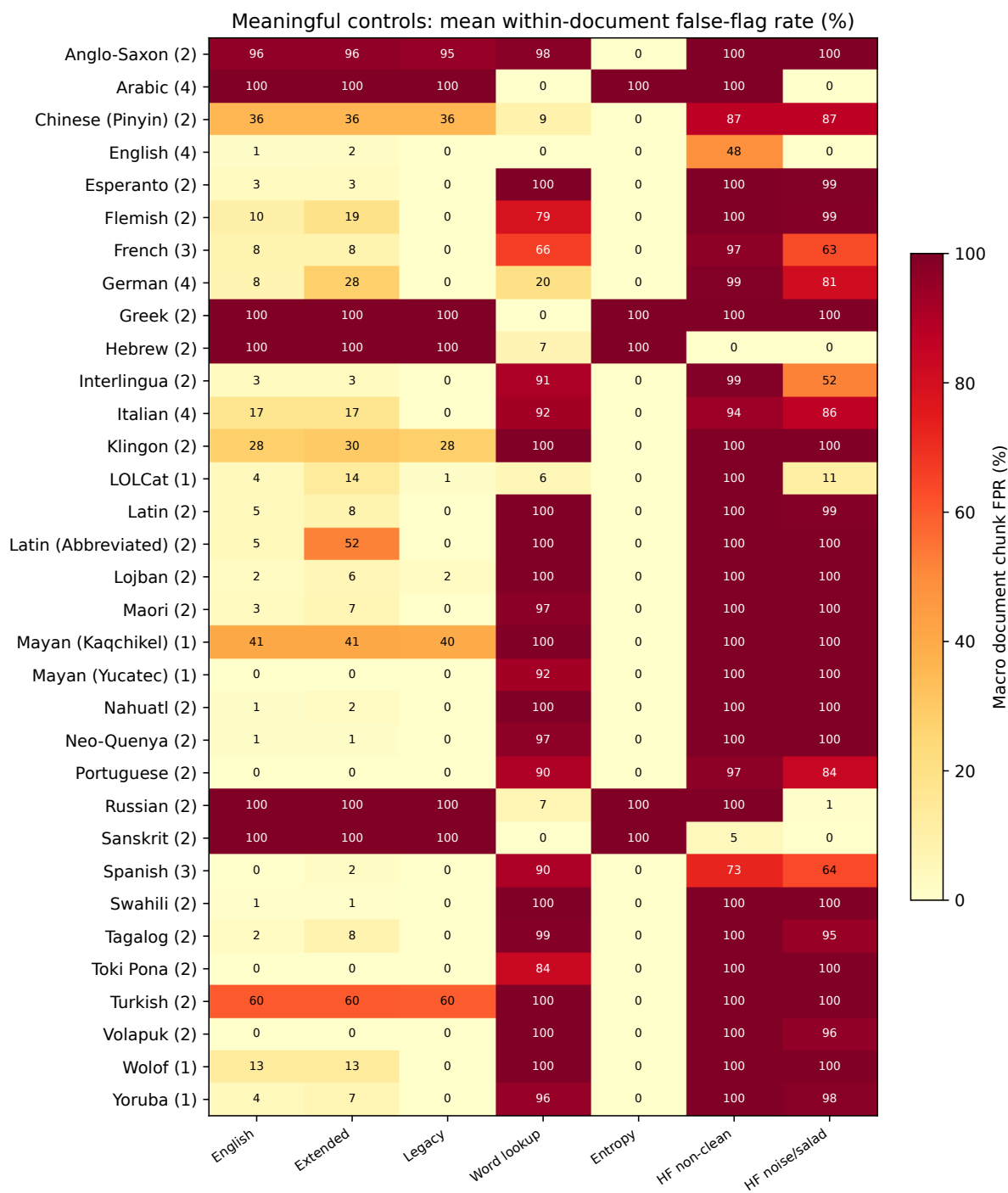


Figure 2: False-flag fractions by supplied language/variant label, averaged over documents. Parentheses show document counts; cells round to whole percentages. Meaningful source labels are retained; English is included for orientation.

CPU and memory cost

The timing workload uses the original fixed 76 inputs of 400 characters: 38 positive prefixes and 38 deterministically selected English control blocks. It is a latency probe, not a random sample of the full multilingual workload. Measurements run on an Apple M2 with 8 GiB RAM, macOS ARM64 and Python 3.12.2, with each method in a fresh process. We use one thread and batch size one for the timing comparison, including tokenizer, model and probability work for HF, and the full score/applicability path for pygarble. After one warm pass, five measured passes produce 380 per-input observations per method. These differ from the four-thread, batched exhaustive run and from pygarble’s potentially faster short-circuit `predict` interface.

Method	Median ms	p95 ms	Process peak MiB
English	0.556	0.673	58.5
Extended	1.048	1.199	50.5
Legacy	0.361	0.440	53.7
Word lookup	0.034	0.040	55.0
Entropy	0.142	0.162	49.9
HF non-clean	36.369	58.842	636.8

Table 5. Warm scoring latency and separate-process peak RSS. Both HF policies share one model run. RSS includes Python, imports, model state and resident corpus data; it is not the isolated model’s tensor memory. Import, construction, first-call and throughput measurements are separately archived.

On this workload, HF median latency is 1057 times word lookup’s and 65.4 times the English profile’s. This describes one CPU, fixed input order and implementation, not a production service-level promise. GPU, quantization, compilation and alternative transformer runtimes were not tested. Model download time is excluded. No energy measurement or cloud-cost extrapolation is made. No paid data, inference endpoint or cloud compute was procured; execution used existing hardware.

Discussion

Full coverage strengthens the earlier comparison by removing its cap on meaningful text and scoring every positive fragment. It also adds a materially different model family and makes its label-policy trade-off visible. The strongest supported claim remains specific: lexical unfamiliarity separates the released invented texts from these four English comparison documents at low measured CPU cost. It does not establish a universal advantage over transformer classifiers. The stricter transformer policy is also effective on the primary task; the broader policy appears mismatched to retaining all inherited meaningful text, particularly the historical *Secreta* source.

A first-stage filter should therefore expose its intended language, rejection policy and fallback behavior. Cheap rejection may be useful when conspicuous lexical noise is the target, but unfamiliar meaningful input can be costly to lose. The full results make those trade-offs inspectable without adding a neural dependency to the library. We have not measured a deployed cascade, real user prevalence, downstream research-quality improvements or independent adoption, and do not infer those benefits from this benchmark.

The preliminary source-held-out calibration experiment remains available with its original protocol and outputs. It fit adapted character bigram/trigram models following Renaud’s approach (Renaud, n.d.) and calibrated thresholds to a 1% empirical validation FPR. Its failures illustrate transfer risk: calibrating word lookup on Wiki yielded 99 false flags among 100 *Secreta* blocks, and the character models produced 42% and 44% Wiki FPR in their held-out fold. These are earlier capped results, not measurements from a new full-corpus calibration run. They motivate caution about carrying a threshold across domains.

Limitations and reproducibility

The most consequential limitation is source diversity. Exhaustive chunking adds text, not independent participants or control domains. The release contains only 38 positive documents and four English controls; participant relationships are not fully known. Meaningful and invented texts were produced under different conditions, and their labels were inherited without a new human audit. Fixed character boundaries and inherited tail labels introduce

further measurement assumptions. The evaluation does not comprehensively cover keyboard noise, ordinary-word semantic nonsense, names, code or modern user submissions.

HF fine-tuning overlap is unknown because its named training dataset could not be inspected. Pretraining and existing package lexical resources also prevent claiming a guaranteed contamination-free test. The earlier and full experiments overlap. The two HF policies and five related package configurations are not seven independent algorithms; small conditional intervals or a zero paired difference cannot establish equivalence. All conclusions are conditional on these inputs and operating rules.

Artifacts include pinned downloads, asset checksums, the frozen scoring-code manifest, per-chunk probabilities and decisions, source-aware summaries and CPU measurements. Raw comparison prose is kept in an ignored local cache. Automated validation checks complete membership and character coverage, recomputes summaries, checks every saved decision mapping and replays a fixed first/middle/last subset from every document in a fresh offline process. This verifies computations, not inherited label validity. The preliminary standard-library study also has a separate byte-identical reproduction. The full transformer run is not claimed to have been independently reproduced across machines or by a human researcher.

Availability and declarations

All study code, provenance, results and manuscript sources are in `paper/study/` of [the pygarble repository](#). `full-results/` holds this evaluation; `results/` retains the preliminary study. The full protocol is `full_corpus_protocol.md`. Exact script/library hashes are recorded in each result directory's code manifest. The optional HF requirements are isolated from the base package.

The author reports using OpenAI Codex and Anthropic Claude in the wider project. Codex assisted with study design, research code, automated execution, analysis and manuscript drafting. No generative LLM produced evaluation labels or served as a judge. The DistilBERT classifier is an explicitly evaluated neural baseline. Claude's precise scope and recoverable model versions remain to be confirmed. Human review of this draft and responsibility for its scientific decisions have not yet been asserted.

Funding and conflicts of interest: author declarations pending. Author review and completed declarations are required before submission. A prospective JOSS article would describe the software contribution and reference this study, rather than duplicate the empirical manuscript. No journal acceptance or search-indexing outcome follows from these measurements.

Appendix: paired recall differences

Method minus HF non-clean	Difference (pp)	Paired 95% interval (pp)
English	-65.8	-78.9 to -50.0
Extended	-47.4	-63.2 to -31.6
Legacy	-65.8	-78.9 to -50.0
Word lookup	0.0	0.0 to 0.0
Entropy	-100.0	-100.0 to -100.0
HF noise/salad	0.0	0.0 to 0.0

Paired document-bootstrap intervals compare the same 38 majority decisions. A zero interval when two methods agree on every document is a property of this empirical resampling distribution, not proof of equal generalization.

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